Supplementary materials

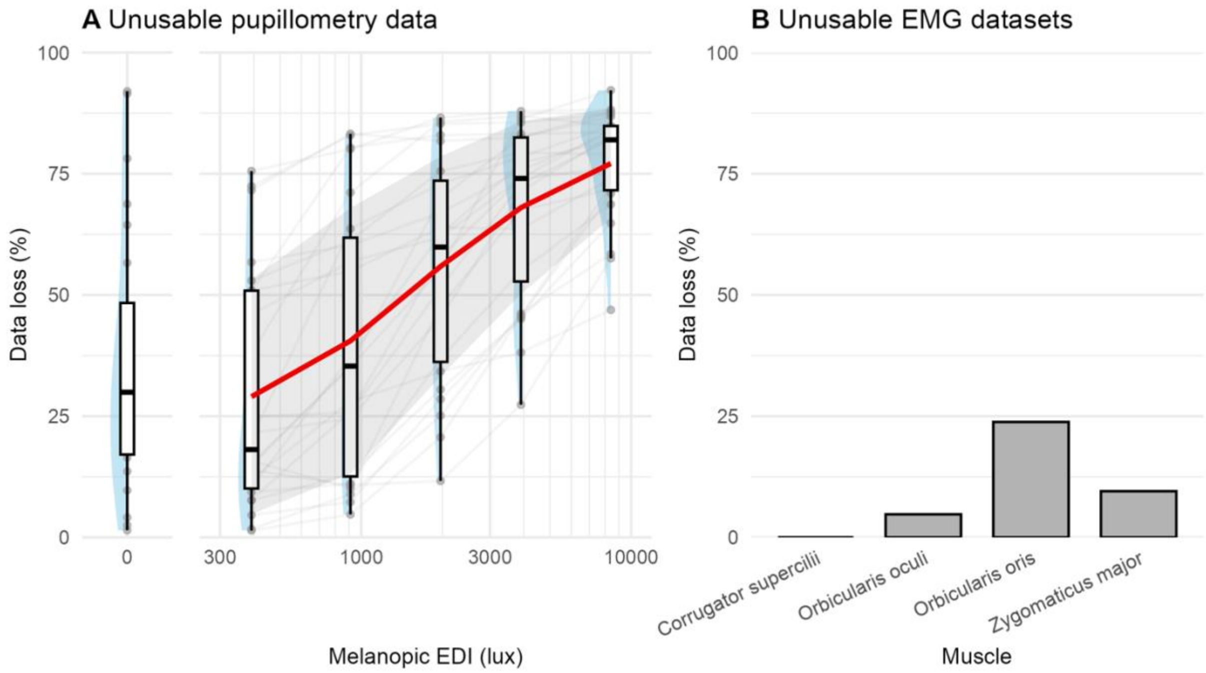

Figure S1. Data loss for pupillometry and EMG modalities, N=18. (A) For pupillometry, data loss is expressed as the proportion of pilot data remaining after filtering out data points with an associated confidence value inferior to a threshold of 0.7. Red line shows group averages and the grey ribbon the standard deviation. (B) For EMG, data loss is expressed as the proportion of pilot data sets that needed to be excluded due to an issue during data collection (electrodes detaching from the skin or computer-related error).

Table S1. Mixed-effects model output for the orbicularis oculi sEMG response to illuminance.

Dependent variable			
Predictors	*Estimates*	*CI*	*p*
(Intercept)	-2.08	-2.25 – -1.91	**<0.001**
Z_i	-0.93	-1.02 – -0.85	**<0.001**
$\log_{10}$(Illuminance)	0.36	0.33 – 0.39	**<0.001**
$\log_{10}$(Baseline EMG)	0.15	0.09 – 0.22	**<0.001**
Trial number	-0.00	-0.00 – -0.00	**<0.001**
Random effects			
σ^2		0.02	
$\tau_{00\ participant\_id}$		0.02	
$\tau_{11\ participant\_id.log\_illuminance}$		0.00	
$\rho_{01\ participant\_id}$		-0.11	
ICC		0.65	
$N_{participant\_id}$		18	
Observations		**648**	
Marginal R^2 / Conditional R^2		**0.445 / 0.807**	

Table S2. Mixed-effects model output for the orbicularis oculi sEMG response to rhodopic EDI.

Dependent variable			
Predictors	*Estimates*	*CI*	*p*
(Intercept)	-2.08	-2.25 – -1.91	**<0.001**
Z_i	-0.89	-0.97 – -0.81	**<0.001**
$\log_{10}$(Rhodopic EDI)	0.35	0.32 – 0.38	**<0.001**
$\log_{10}$(Baseline EMG)	0.16	0.09 – 0.22	**<0.001**
Trial number	-0.00	-0.00 – -0.00	**<0.001**
Random effects			
σ^2		0.02	
$\tau_{00\ \mathrm{participant\_id}}$		0.02	
$\tau_{11\ \mathrm{participant\_id.log\_rhodopic\_edi}}$		0.00	
$\rho_{01\ \mathrm{participant\_id}}$		-0.11	
ICC		0.65	
N $_{\mathrm{participant\_id}}$		18	
Observations		**648**	
Marginal R^2 / Conditional R^2		**0.444 / 0.806**	

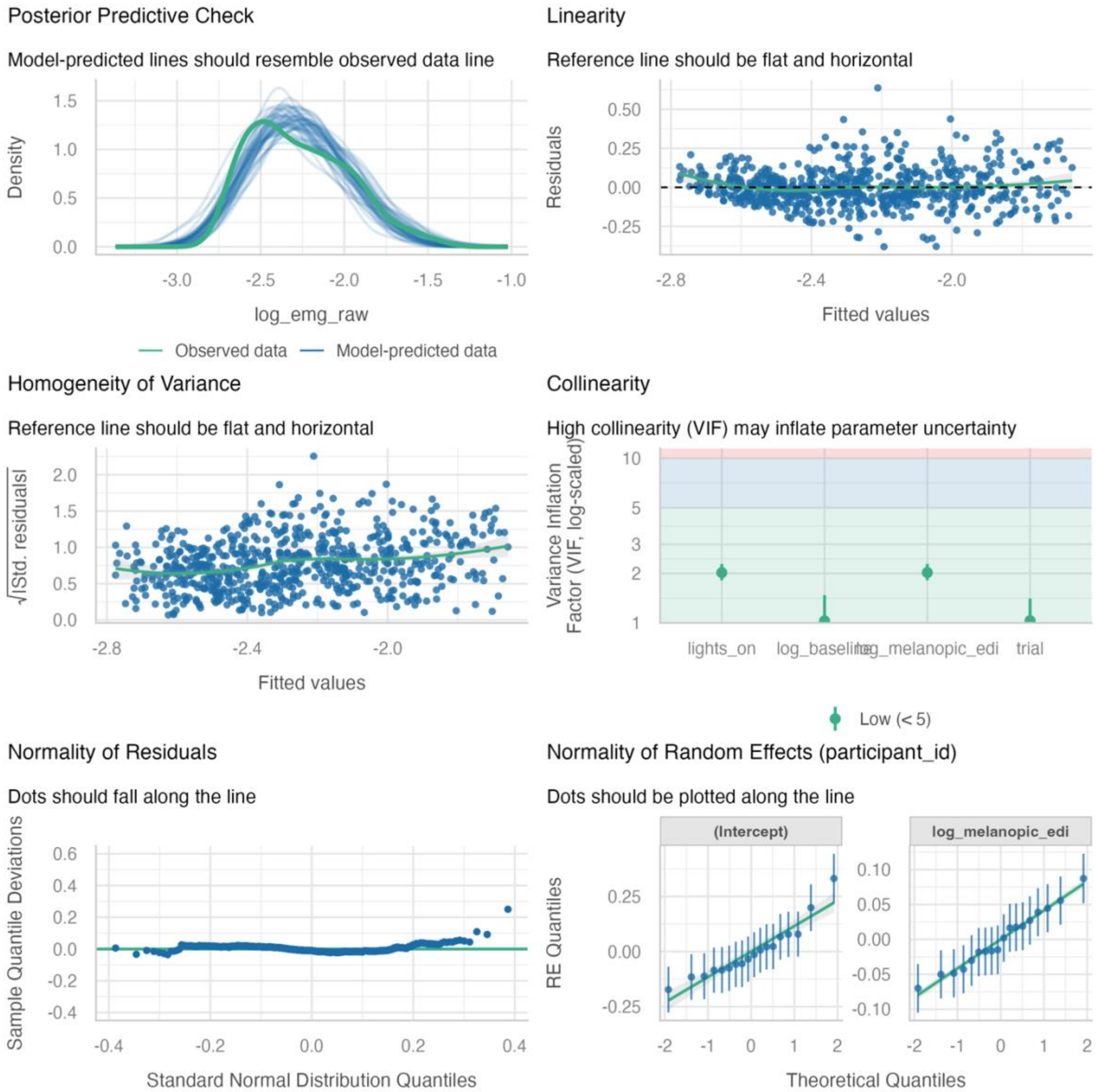

Figure S2. Model diagnostics for the mixed-effects model for the orbicularis oculi sEMG response to melanopic EDI fit on pilot data, obtained using the check_model() function of the performance R package.

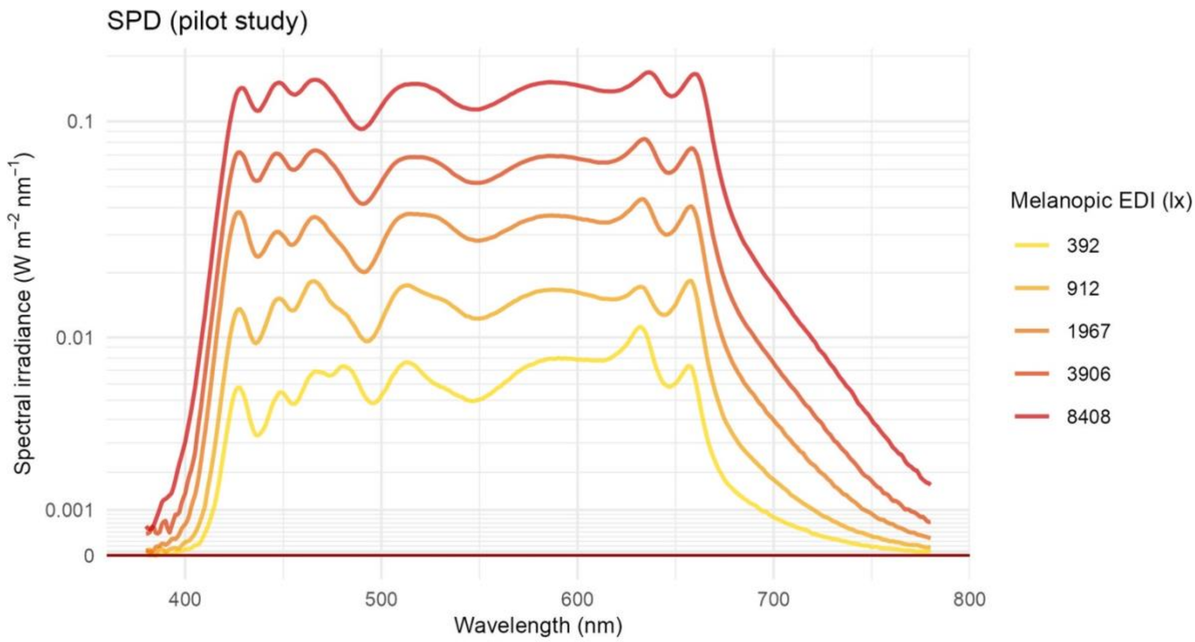

Figure S3. Spectral power distributions (SPD) of pilot study light stimuli. The plot shows absolute spectral irradiance (380-780 nm) of the primary light settings used in the pilot study (melanopic EDI = 392, 912, 1967, 3906 and 8408 lx), excluding the 0-lx condition. The y-axis was transformed using a base-10 pseudo-logarithmic transformation with a linear region around zero (σ = 0.001).

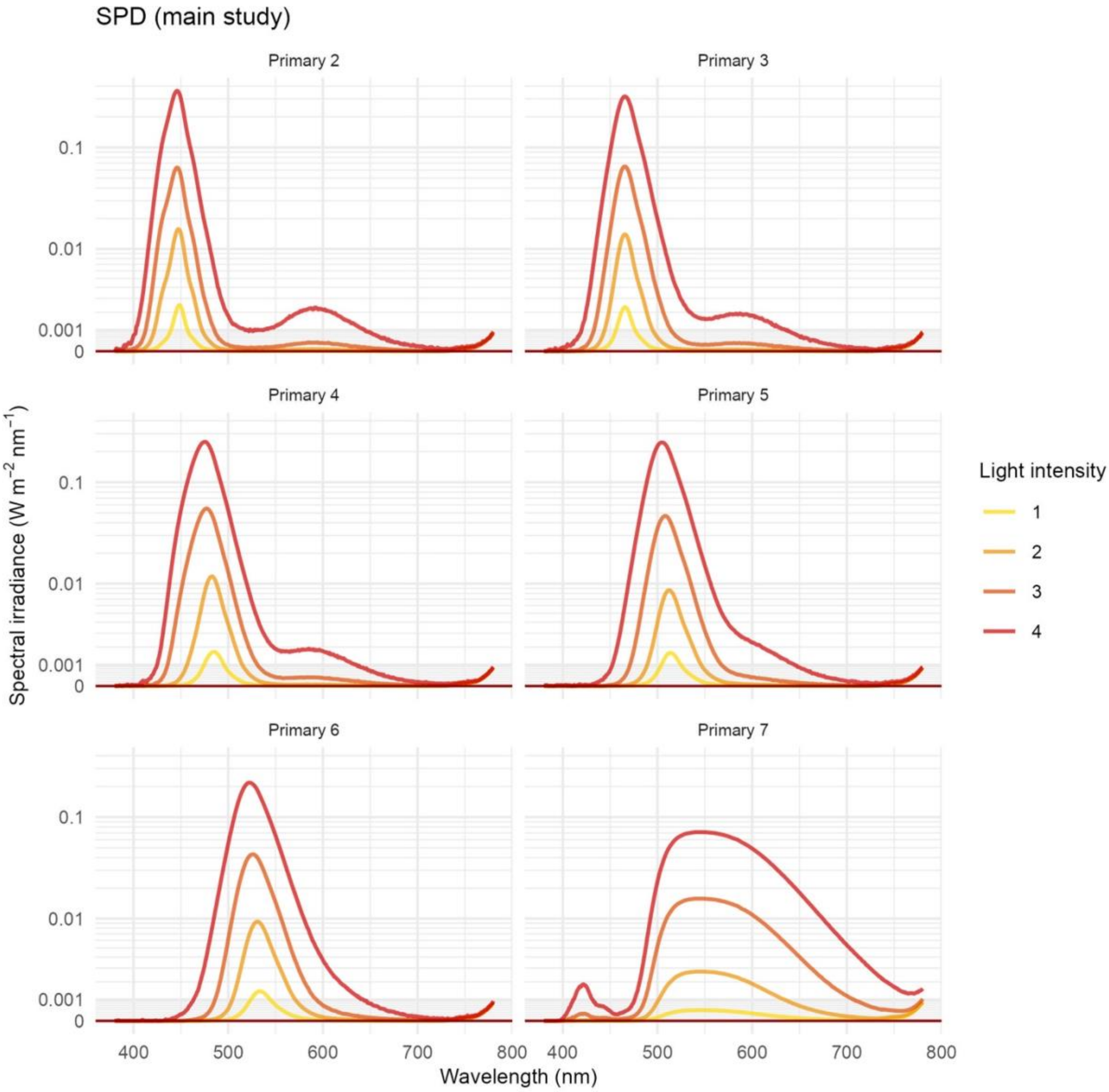

Figure S4. Spectral power distributions (SPD) of pilot study light stimuli. The plots show absolute spectral irradiance (380-780 nm) of the primary light settings used in the main study, for each primary (2 to 7) and each irradiance setting (0.06, 0.32, 1.6, 8.8 $W.m^{-2}$). The y-axis was transformed using a base-10 pseudo-logarithmic transformation with a linear region around zero (σ = 0.001). The measurements were made with the infrared camera at the back of the integrating sphere on, which introduces radiation in the infrared range.

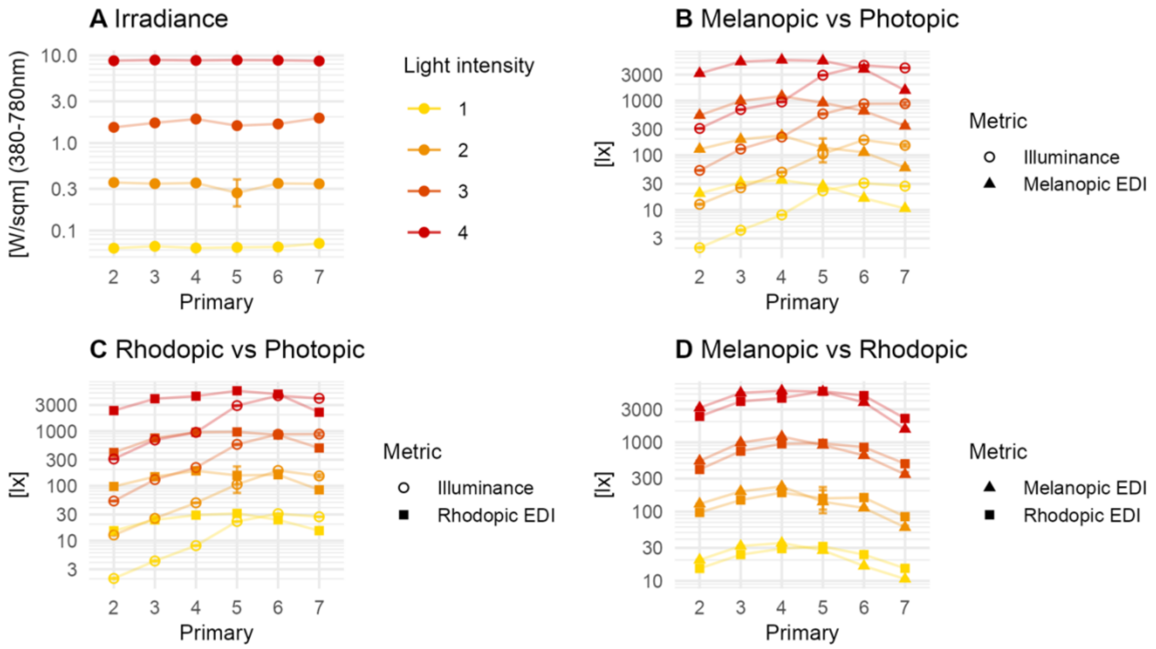

Figure S5. Calibration measurements of the light stimuli, across all possible light intensities that participants will get randomly during each of their laboratory visits. (A) Irradiance stays constant across primaries, and so for each light intensity setting. (B) Primaries with peaks at lower wavelengths show higher melanopic excitation (melanopic EDI, tuned to ipRGCs) than photopic excitation (illuminance, tuned to L+M cones) than primaries at higher wavelengths. As the wavelength increases, the trend reverses – photopic excitation increases and surpasses melanopic excitation. (C) Primaries with peaks at lower wavelengths show higher rhodopic excitation (rhodopic EDI, tuned to rods) than photopic excitation than primaries at higher wavelengths, and vice-versa. (D) In the standard human observer, ipRGCs and rods have very close peak sensitivities, respectively ~507 and ~480 nm. Consequently, the design will tend to produce near-parallel stimulation of ipRGC- and rod-weighted pathways; this is an inherent limitation that the present design cannot independently compensate for.

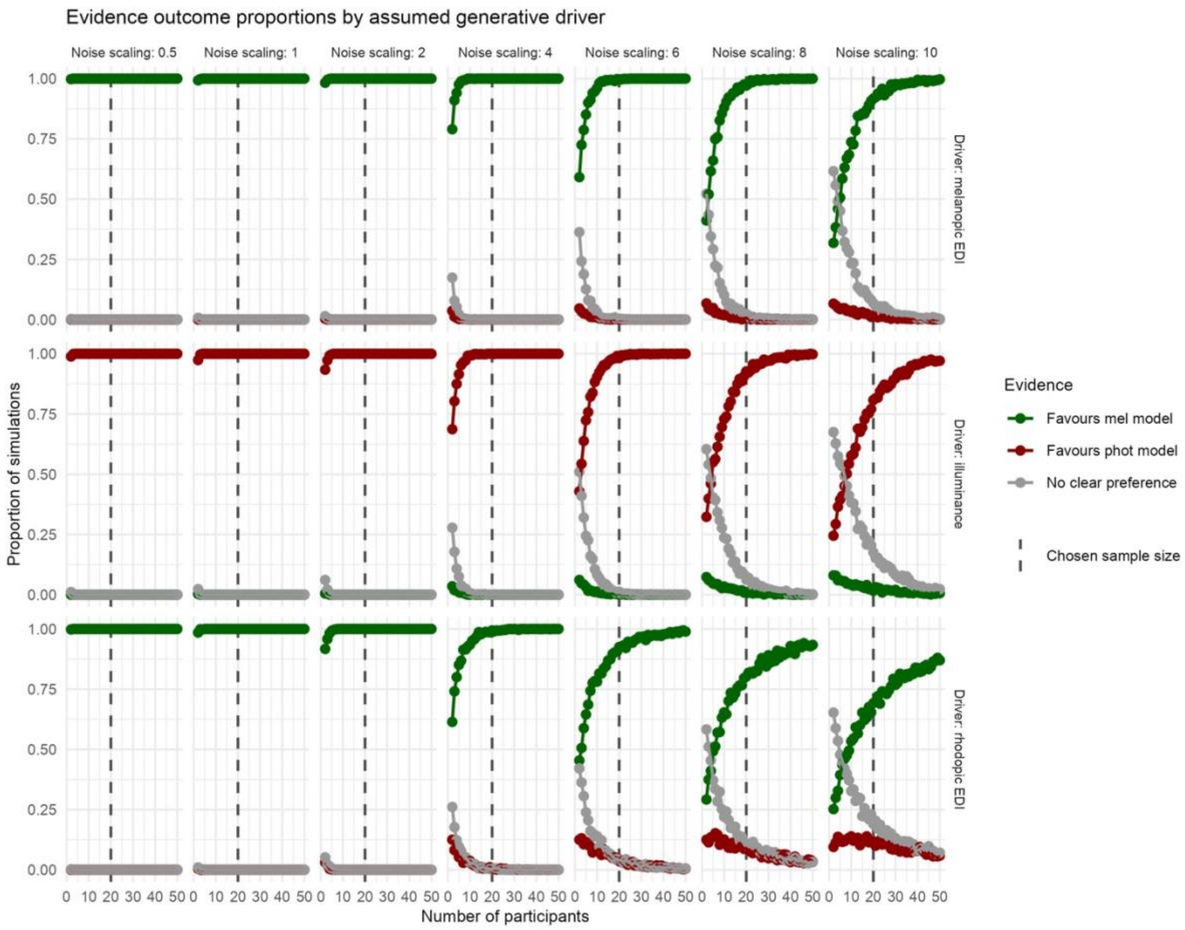

Figure S6. (A) Evidence outcome proportions depending on sample size (2 to 50 participants), residual noise scaling level (0.5x, 1x, 2x, 4x, 6x, 8x and 10x the residual noise level from the generative model) and assumed generative driver (melanopic, photopic, or rhodopic), using AIC-based model comparisons on data simulated using pilot data. The dashed black line indicates the selected sample size of n=20. For each combination of factors, 1000 simulations are run to compute the proportions. The results show that, for our chosen sample size of n=20, even for noise levels higher than what was observed in the pilot data, the design is able to successfully recover the driving response from simulated data. Given the overlap between rhodopic and melanopic curves, the design naturally recovers a melanopic driver when we assume a rhodopic driver.

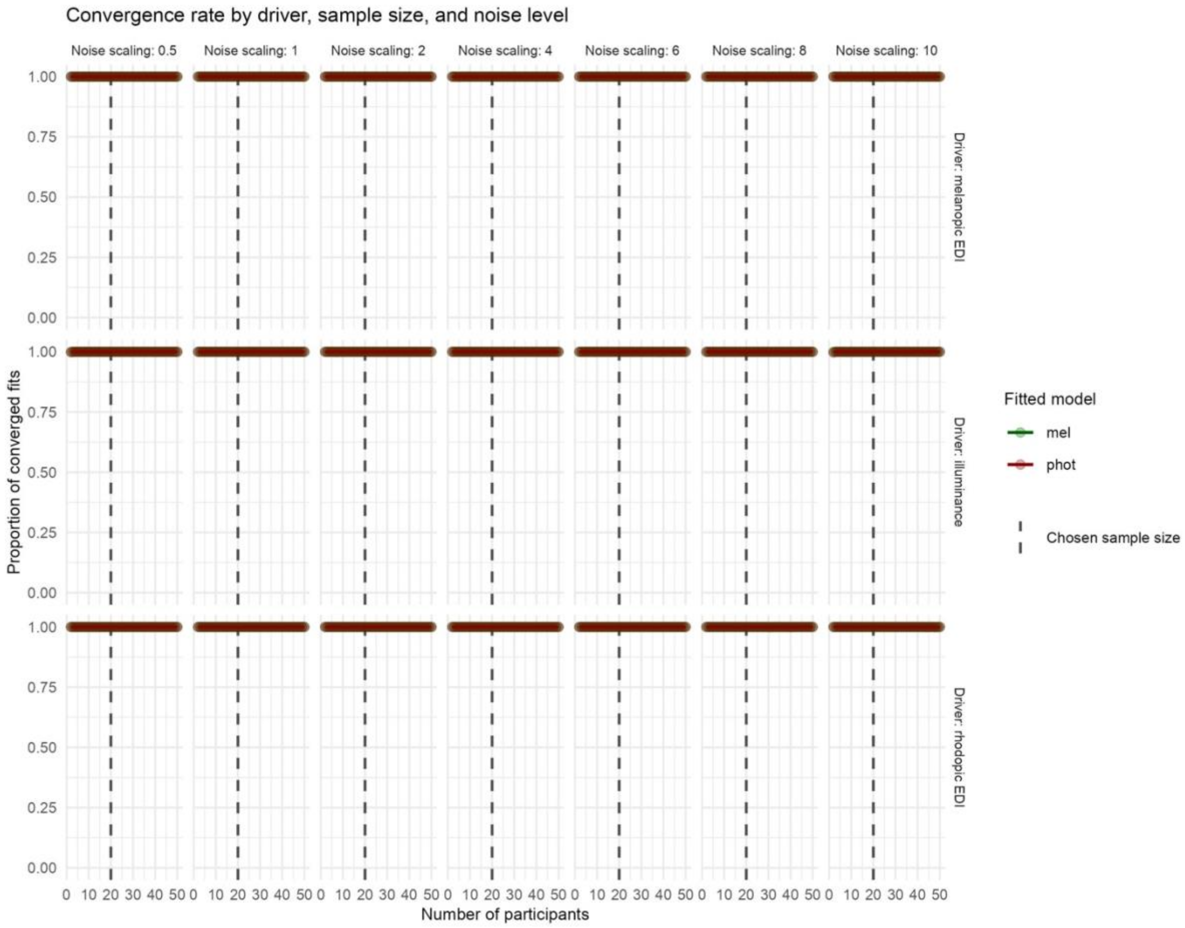

Figure S7. Linear mixed model convergence rates depending on sample size (2 to 50 participants), residual noise scaling level (0.5x, 1x, 2x, 4x, 6x, 8x and 10x the residual noise level from the generative model) and assumed generative driver (melanopic, photopic, or rhodopic), associated with the evidence outcome proportions reported in Figure S5. Convergence rates are consistently near or equal to 1, i.e. no convergence warnings nor singular fits. This indicates that the proposed analysis pipeline is numerically stable for the planned design and that subsequent model-comparison estimates are unlikely to be driven by fitting failures.

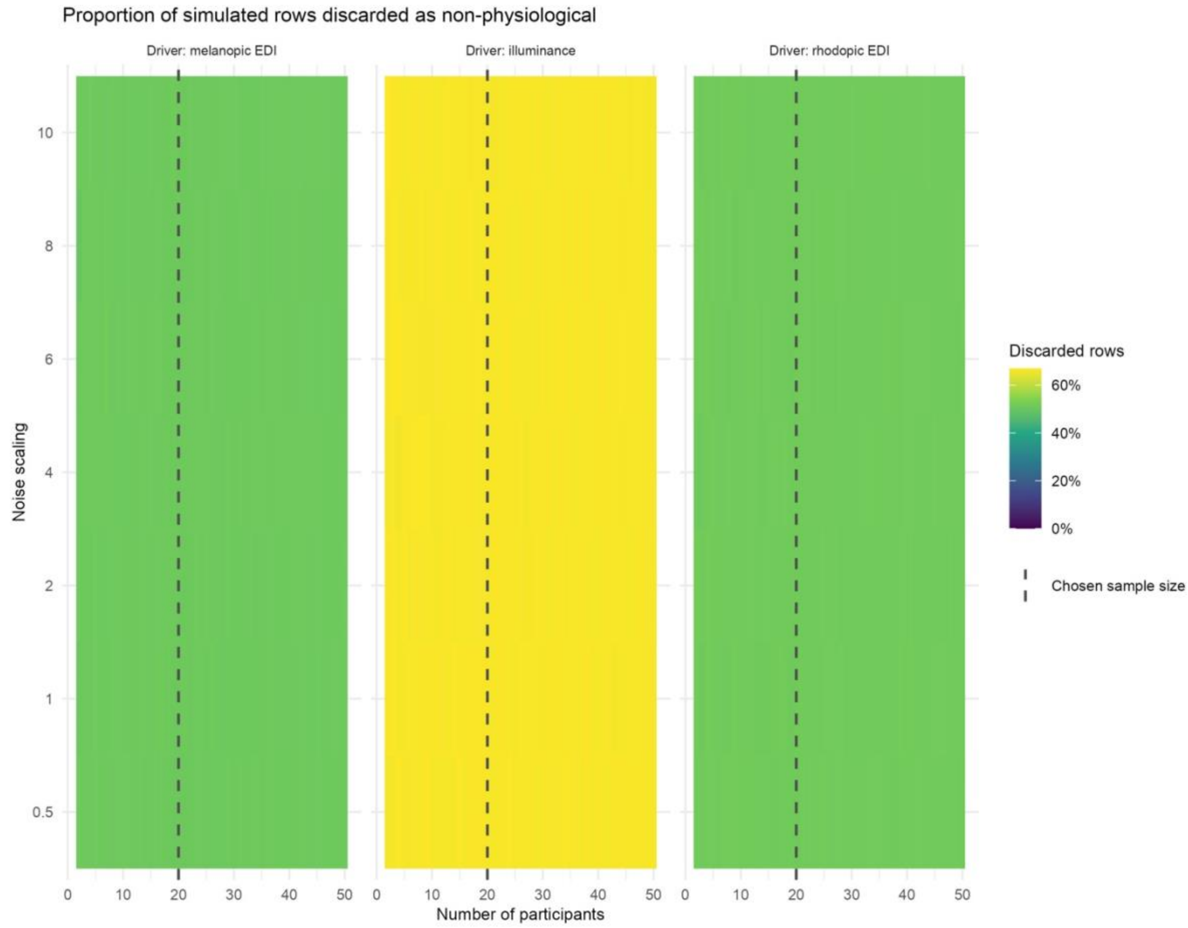

Figure S8. Proportion of simulated data remaining after discarding non-physiological values, i.e. values below the observed response at 0 lx in the pilot data. For a melanopic and rhoodpic driver, the proportion is respectively of 51.0 % and 51.6 % on average, and for a photopic driver, the proportion goes up to 66.5 %. This is to be expected as the design focuses more on short-wavelength stimulation which therefore leads to more data falling under the physiological limit. No dependence on sample size or noise level can be observed.

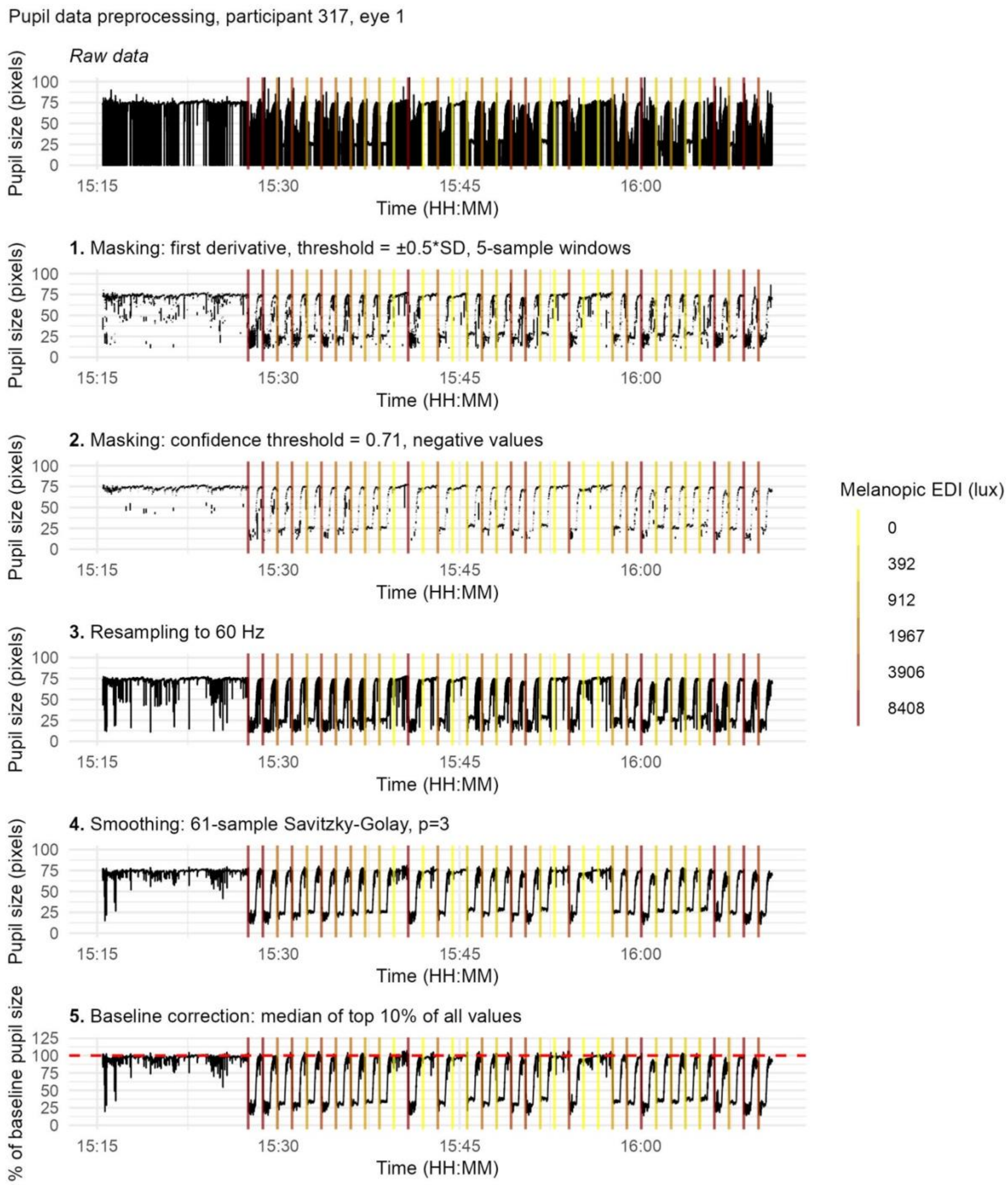

*Figure S9. Pupillometry data preprocessing pipeline, for an example data set from the pilot data. Panels show (from top to bottom): raw pupillary signal; masking based on the first derivative with a threshold of ±0.5*SD to the local mean for 5-sample windows; resampling to 60 Hz; masking based on confidence threshold, defined programmatically as the 60% quantile of all confidence values in the dataset, with boundaries set from 0.6 to 0.71, and masking based on physiologically impossible values, i.e. negative values, resampling to 60 Hz, smoothing using a third-order polynomial Savitzky-Golay filter with 61-sample windows; and baseline-corrected pupil size expressed as a percentage of baseline, corresponding to the median of the top 10% values of the whole recording used for normalisation. Vertical lines indicate light onset times, colour-coded by melanopic equivalent daylight illuminance (EDI). All panels share a common time axis.*

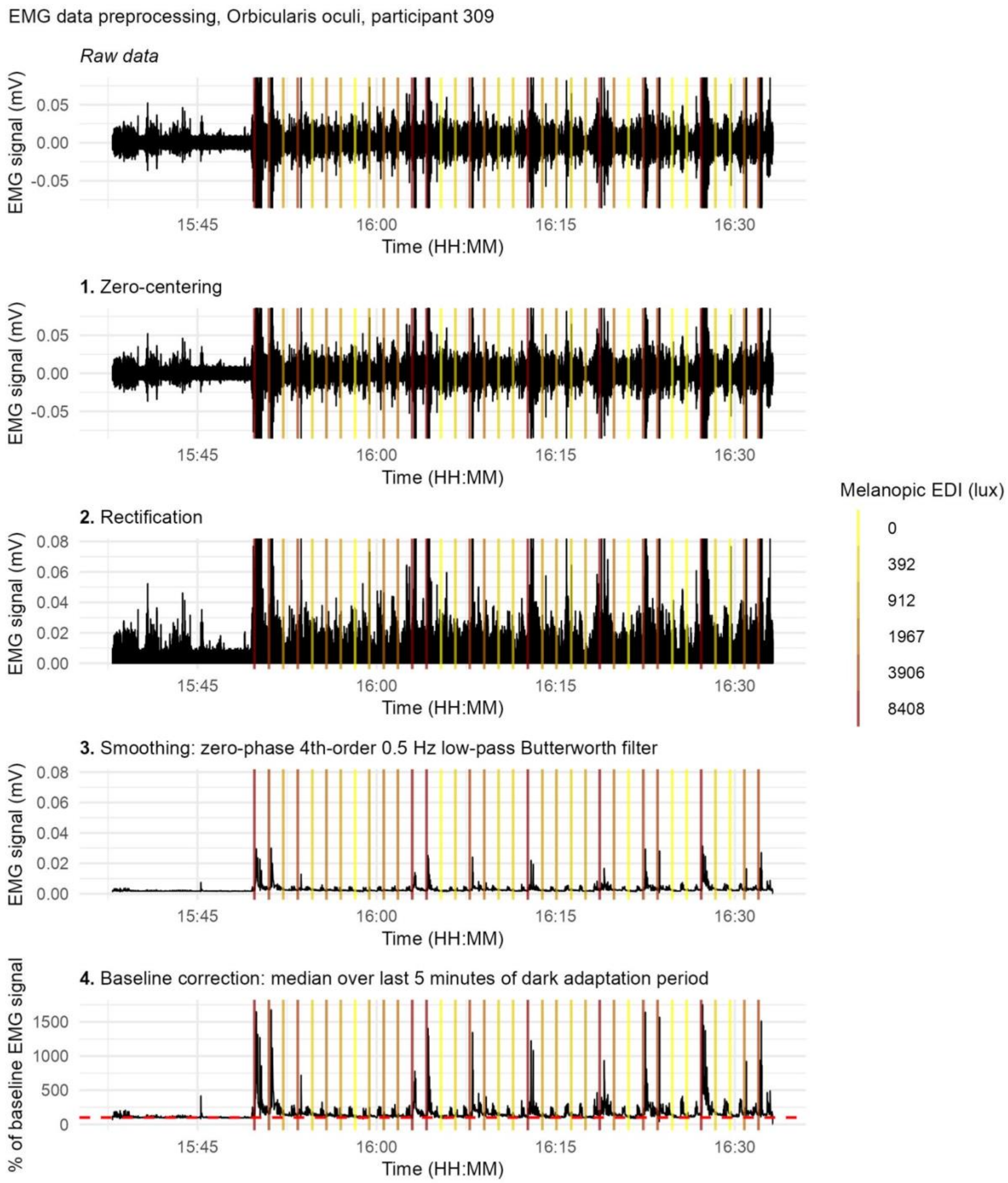

Figure S10. sEMG data signal preprocessing pipeline, for an example data set from the pilot data. Panels show (from top to bottom): raw EMG signal; zero-centred signal; rectified signal; smoothed signal obtained via zero-phase forward-backward fourth-order Butterworth low-pass filtering at 0.5 Hz; and baseline-corrected EMG expressed as a percentage of baseline, corresponding to the final five minutes of the dark adaptation period used for normalisation. Vertical lines indicate light onset times, colour-coded by melanopic equivalent daylight illuminance (EDI). All panels share a common time axis, and y-axis limits are constrained to the upper 99.99th quantile of the data to improve visualisation (removing frequent extreme outliers) while preserving relative signal dynamics.

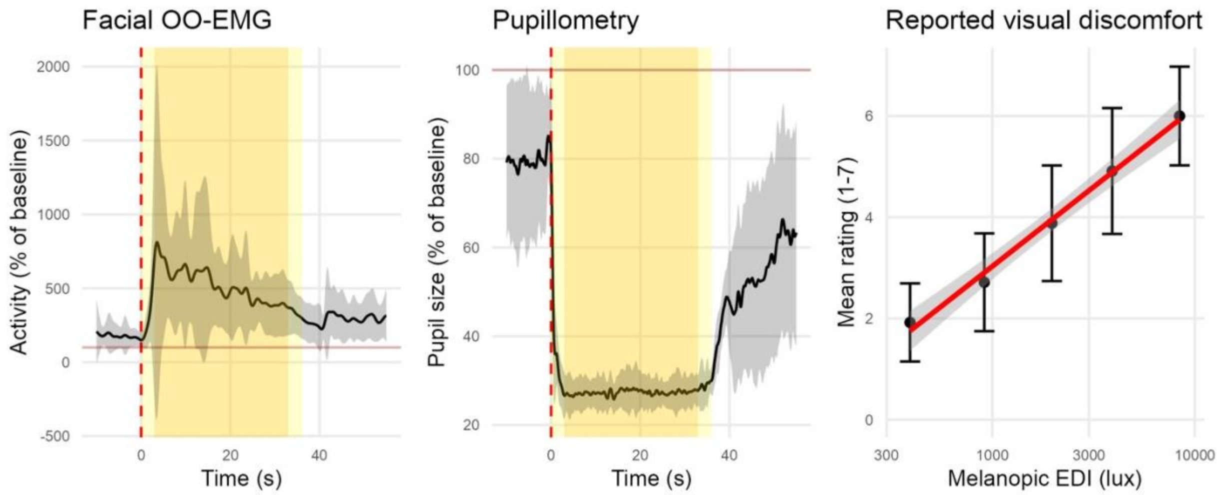

Figure S11. Pilot data, showing group averages, N=18. Panels show (from left to right): trace obtained for orbicularis oculi sEMG activity from 10 seconds before light onset to 30 seconds after light offset; typical trace obtained for pupil size from 10 seconds before light onset to 30 seconds after light offset, and self-reported visual discomfort ratings as a function of light intensity (melanopic EDI).